

P R O J E C T P R O P O S A L

Medical Store Management System

A Comprehensive Database Design & Implementation

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A B S T R A C T

This proposal presents the design and implementation of a Medical Store Management System using MySQL. The system aims to digitize and streamline all core operations of a medical store, including inventory management, sales processing, prescription tracking, supplier coordination, and employee management. The proposed database comprises 12 interrelated tables with advanced features such as triggers, stored procedures, views, and transactions, ensuring data integrity, efficiency, and real-world applicability.

1. Introduction

In Pakistan and across the world, medical stores serve as a critical link between pharmaceutical manufacturers and patients. Managing such a store manually is prone to errors, stock mismanagement, and regulatory non-compliance. A well-designed database system can eliminate these issues by providing accurate, real-time information at every stage of the store's operations.

The Medical Store Management System (MSMS) proposed here is a relational database project built using MySQL. It is designed to handle all day-to-day activities of a modern medical store, from tracking medicine inventory and supplier orders to managing customer prescriptions and generating sales reports.

2. Objectives

The primary objectives of this project are:

- Design a normalized relational database (up to 3NF) for a medical store environment.
- Implement all core MySQL features including DDL, DML, JOINS, subqueries, views, stored procedures, triggers, and transactions.
- Ensure data integrity through proper use of primary keys, foreign keys, and constraints.
- Simulate real-world business logic such as automatic stock reduction on sale and low-stock alerts.
- Provide meaningful reports through SQL views and queries for management decision-making.
- Demonstrate proficiency in database design as required by the Database Lab course.

3. Project Scope

The scope of this project covers the following functional modules:

#	Module	Description
1	Inventory Management	Track medicines, quantities, batch numbers, and expiry dates
2	Sales & Orders	Process customer orders, generate bills, apply discounts
3	Prescription Tracking	Link prescriptions from doctors to customer purchases
4	Supplier Management	Manage supplier records, ratings, and stock deliveries
5	Customer Management	Maintain customer profiles and purchase history
6	Employee Management	Track pharmacists, cashiers, and their roles
7	Payment Processing	Record payments via cash, card, online, or insurance
8	Reporting & Analytics	Sales reports, expiry alerts, and revenue summaries via views

4. Proposed Database Schema

The database will consist of 12 tables. Each table is designed to eliminate data redundancy and ensure referential integrity. The tables and their primary responsibilities are outlined below:

Table Name	Primary Key	Role / Purpose
Categories	category_id	Classifies medicines (Antibiotic, Painkiller, etc.)
Manufacturers	manufacturer_id	Stores pharmaceutical company information
Suppliers	supplier_id	Tracks suppliers with ratings and contact info
Medicines	medicine_id	Core medicine catalog with dosage, price, prescription flag
Stock	stock_id	Tracks quantity, batch, manufacturing and expiry dates
Doctors	doctor_id	Stores licensed doctor details for prescriptions
Customers	customer_id	Customer profiles with demographics and contact info
Employees	employee_id	Staff records including role, salary, and hire date
Prescriptions	prescription_id	Links doctors and customers; validates prescription-only sales
Orders	order_id	Master order record with status, discount, and total
Order_Items	item_id	Line items per order with computed subtotal column
Payments	payment_id	Tracks payment method, status, and transaction reference

5. Entity Relationships

The following key relationships exist between entities in the database:

- A Medicine belongs to one Category and is produced by one Manufacturer.
- Each Medicine has one Stock record linked to a Supplier.
- A Customer can have multiple Orders and multiple Prescriptions.
- Each Prescription is issued by a Doctor and may be linked to an Order.
- An Order contains multiple Order_Items, each referencing a Medicine.
- Each completed Order has exactly one Payment record.
- An Employee processes each Order (acting as cashier or pharmacist).

6. Advanced MySQL Features to be Implemented

Feature	Implementation Plan
TRIGGER	Auto-decrement stock quantity when an Order_Item is inserted
TRIGGER	Raise an alert flag when stock quantity falls below reorder level
STORED PROCEDURE	GenerateBill(order_id) -- calculates total with discount and tax
STORED PROCEDURE	RestockMedicine(medicine_id, qty) -- adds quantity to stock

Feature	Implementation Plan
VIEW	vw_LowStockAlert -- shows medicines where quantity < reorder level
VIEW	vw_ExpiringSoon -- lists medicines expiring within 30 days
VIEW	vw_SalesSummary -- daily/monthly revenue grouped by date
VIEW	vw_TopSellingMedicines -- most sold medicines by quantity
TRANSACTION	Wrap order placement + payment + stock update in a single transaction
INDEX	Optimize queries on medicine_name, expiry_date, and customer phone

7. Normalization

The database will be normalized to Third Normal Form (3NF):

- 1NF: All tables have atomic columns with no repeating groups. Each row is uniquely identifiable by its primary key.
- 2NF: All non-key attributes are fully functionally dependent on the entire primary key. Composite keys are avoided by using surrogate integer keys.
- 3NF: No transitive dependencies exist. For example, Manufacturer details are separated from Medicines to avoid storing country and email redundantly for each medicine.

8. Tools & Technologies

Tool / Technology	Purpose
MySQL 8.0	Primary RDBMS for database creation and query execution
MySQL Workbench	GUI for schema design, ER diagram generation, and query testing
draw.io / Lucidchart	ER Diagram and relational schema visualization
VS Code	SQL file editing and project documentation
Microsoft Word	Project report and proposal documentation

9. Project Timeline

Week	Task
Week 1	Finalize requirements, design ER diagram, identify all entities and relationships
Week 2	Create all 12 tables with constraints, insert sample data (10+ rows per table)
Week 3	Write all SELECT queries, JOINS, subqueries, GROUP BY, and HAVING clauses
Week 4	Implement Triggers, Stored Procedures, Views, Indexes, and Transactions
Week 5	Testing, debugging, final documentation, and submission

10. Conclusion

The Medical Store Management System is a well-scoped, practically relevant database project that will demonstrate a thorough understanding of relational database concepts. By covering all major MySQL features including DDL, DML, constraints, JOINS, views, triggers, stored procedures, transactions, and normalization, this project fulfills all requirements of the Database Lab course.

The system design reflects real-world scenarios encountered in pharmacy management, making it both academically rigorous and practically valuable. Upon completion, the deliverables will include a complete .sql file, ER diagram, relational schema, and a comprehensive project report.

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